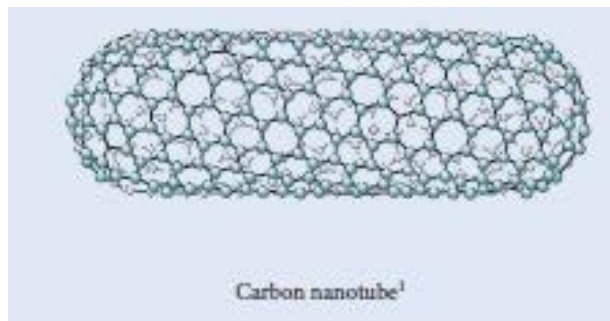
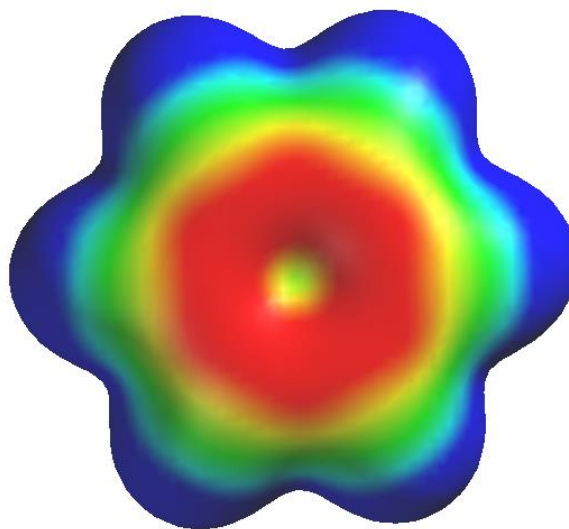


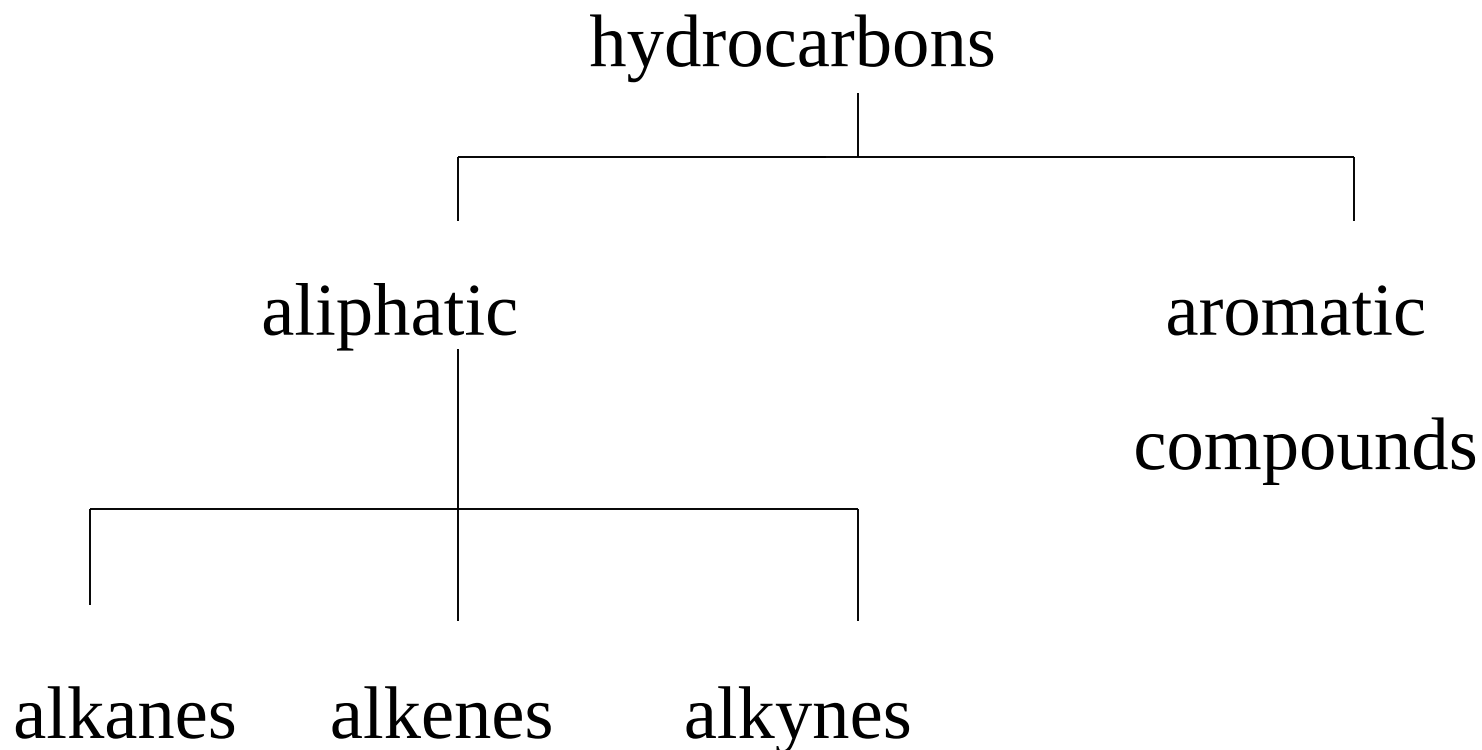


Chapter 7

Aromatic Compounds



Aromaticity

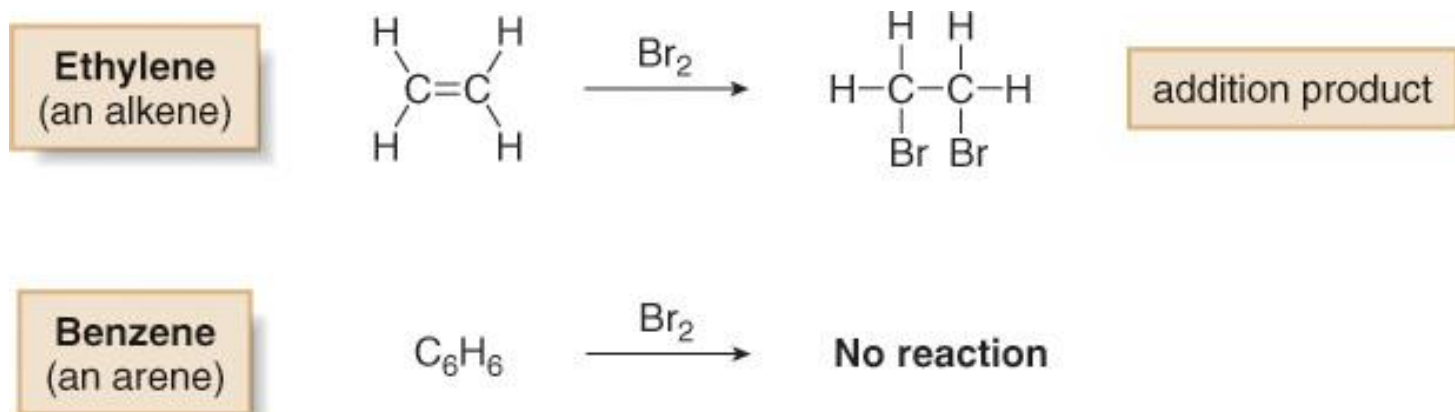


Aromatic compounds: unsaturated ring compounds that are more stable than they should be and resist the addition reactions typical of unsaturated aliphatic compounds.

Benzene and related compounds.

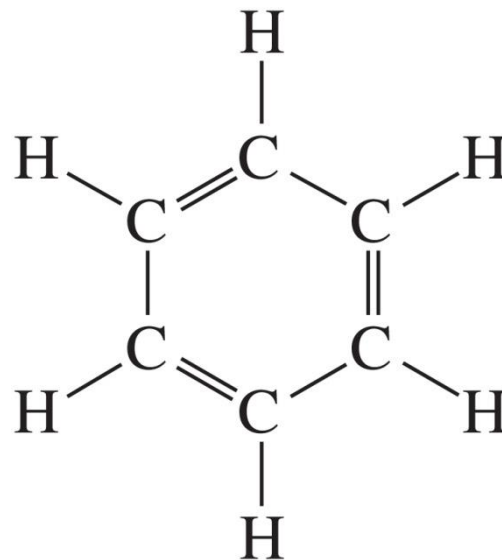
Benzene

- Benzene (C_6H_6) is the simplest aromatic hydrocarbon
- Benzene is a highly unsaturated hydrocarbon.
- Whereas unsaturated hydrocarbons such as alkenes, alkynes and dienes readily undergo addition reactions, benzene does not.



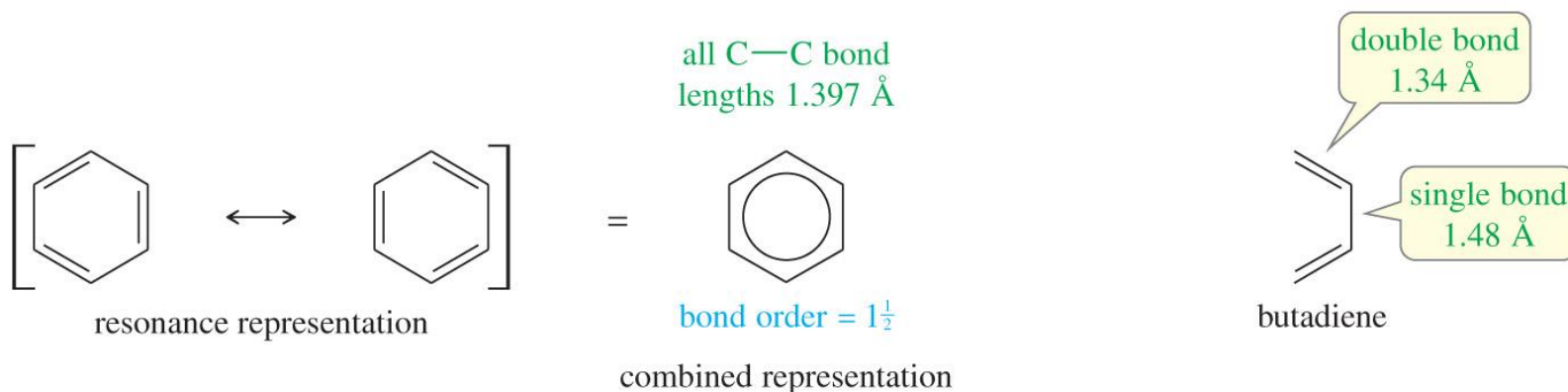
4.2 The Kekulé Structure of Benzene

- Proposed in 1866 by Friedrich Kekulé, shortly after multiple bonds were suggested.



Kekulé structure
of benzene

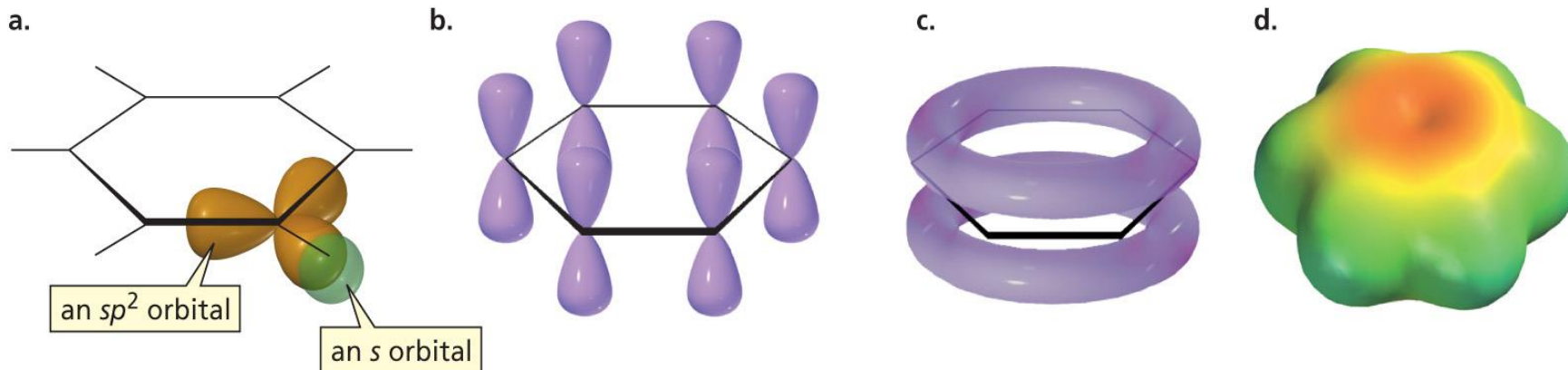
4.3 Resonance Model for Benzene



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- Benzene is actually a resonance hybrid between the two Kekulé structures.
- The C—C bond lengths in benzene are shorter than typical single-bond lengths, yet longer than typical double-bond lengths (bond order 1.5).
- Benzene's resonance can be represented by drawing a circle inside the six-membered ring as a combined representation.

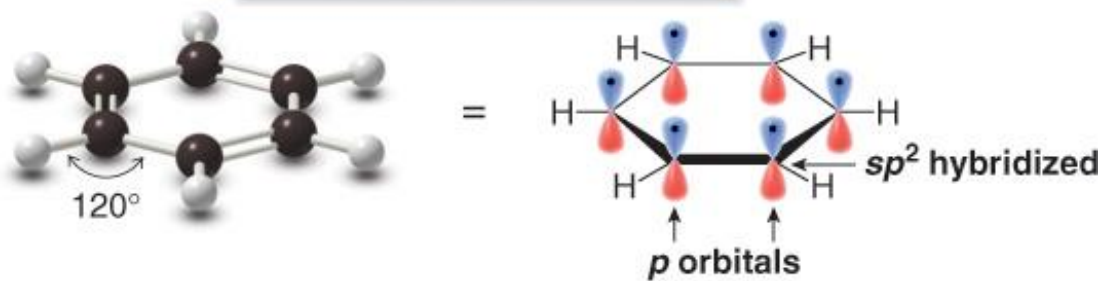
4.4 Orbital Model for Benzene



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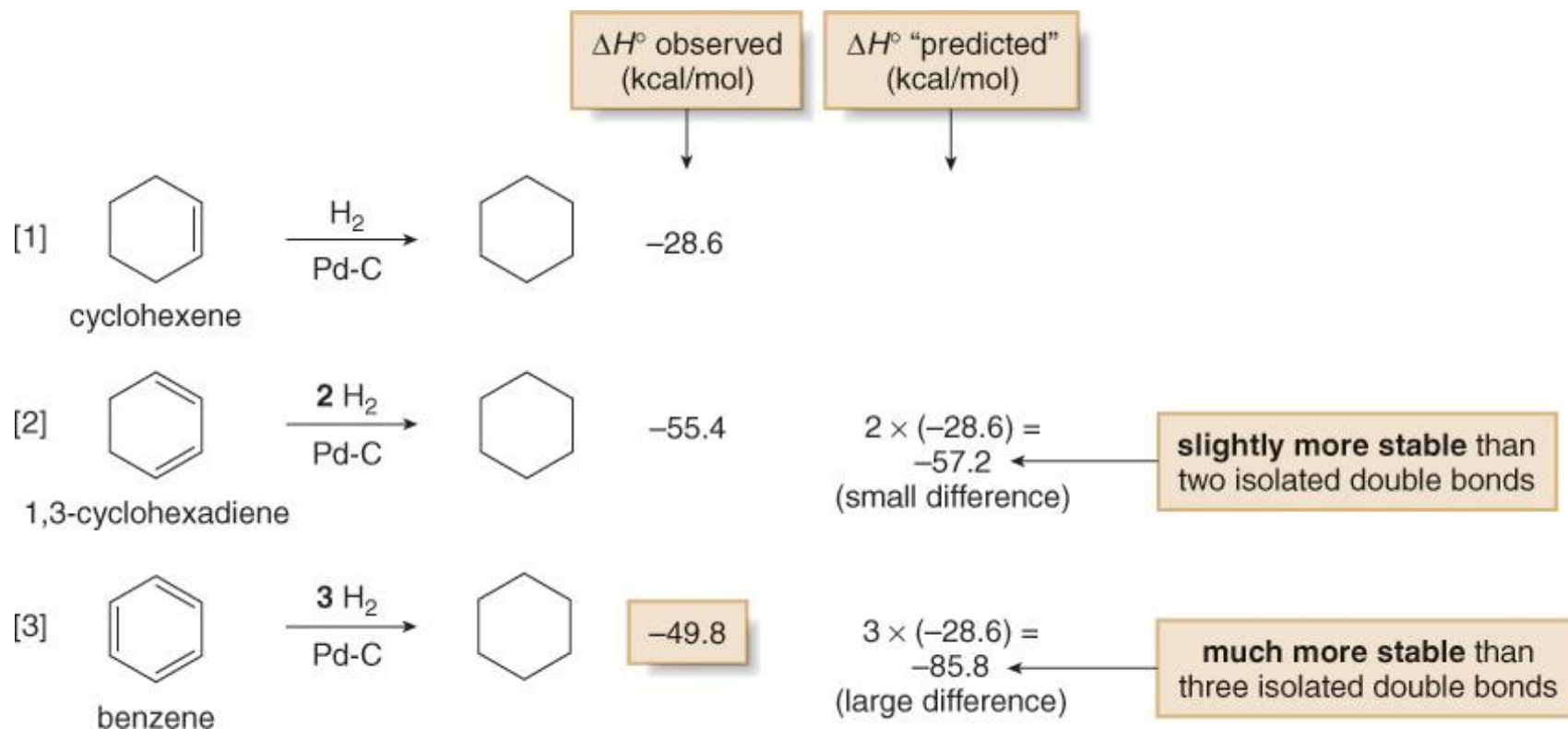
- Benzene is planar, and each carbon atom is at the corner of a regular hexagon. All of the
- carbon–carbon bond lengths are identical: 1.39
- Each sp^2 hybridized C in the ring has an unhybridized p orbital perpendicular to the ring that overlaps around the ring.
- The six pi electrons are delocalized over the six carbons.

Benzene—A planar molecule



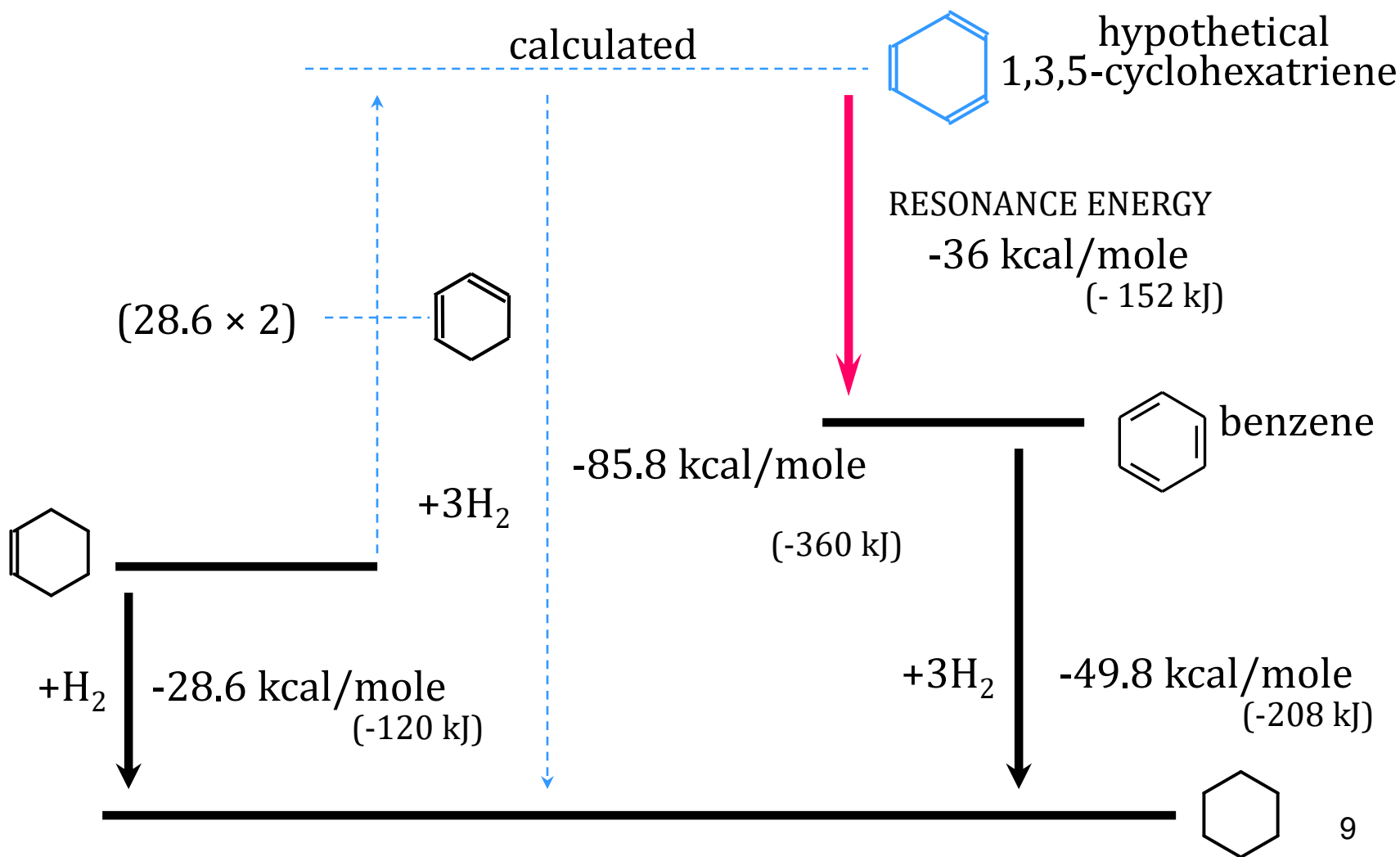
4.7 The Resonance Energy of Benzene

- Consider the heats of hydrogenation of cyclohexene, 1,3-cyclohexadiene and benzene, all of which give cyclohexane when treated with excess hydrogen in the presence of a metal catalyst.



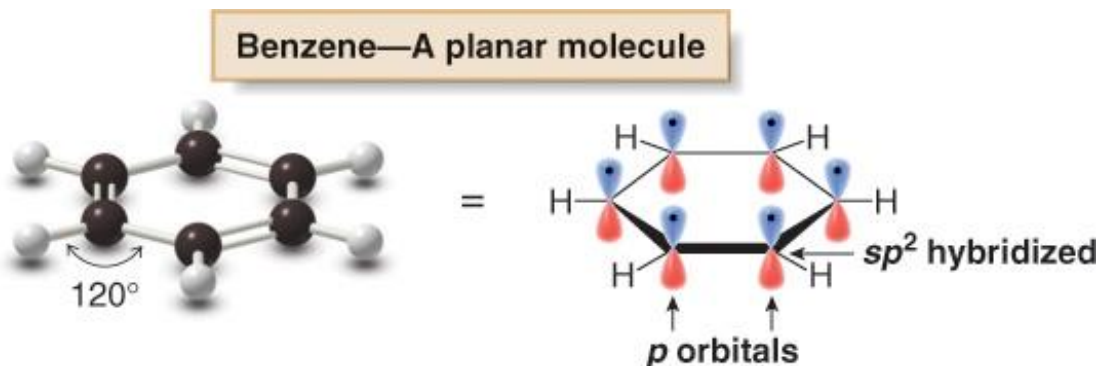
The **stabilization energy, or resonance energy, of a substance** is the difference between the energy of the real molecule and the calculated energy of the most stable contributing structure.

Figure 4.3 Comparison of heat released by the hydrogenation of cyclohexene, 1,3-cyclohexadiene, and benzene to produce cyclohexane.



Criteria for Aromaticity

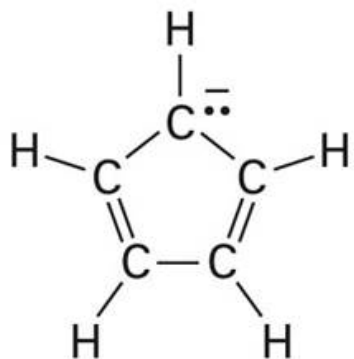
- 1- the molecules must be cyclic
- 2- all the atoms in the ring has sp^2 hybridization to form conjugated double bonds
- 3- the molecule must be planar
- 4- number of π electrons $= 4n + 2$ (n is 0,1,2,3,4)



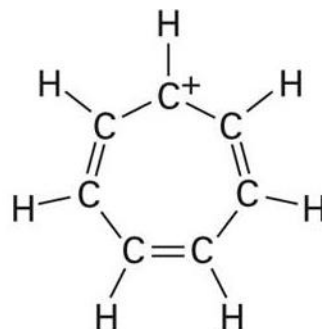
Number of π electrons in the aromatic compounds

n	$4n + 2$
0	2
1	6
2	10
3	14
4, etc.	18

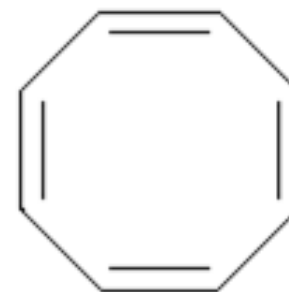
Aromaticity



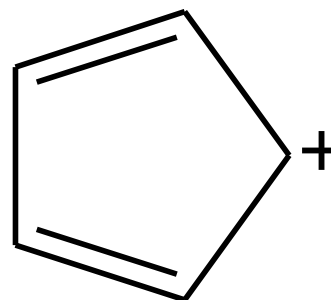
Cyclopentadienyl anion



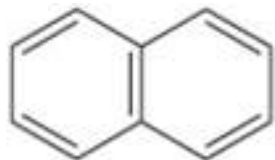
Cycloheptatrienyl cation



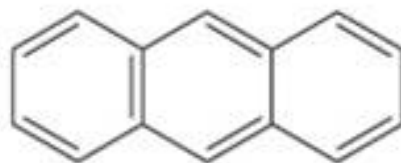
← no *p* orbital



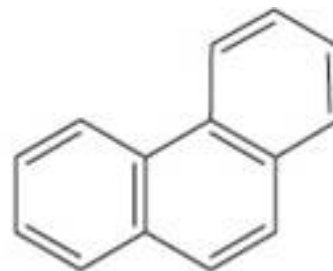
Polycyclic Aromatic Compounds



naphthalene
10 π electrons

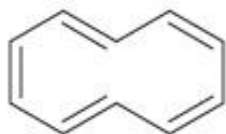


anthracene
14 π electrons



phenanthrene
14 π electrons

[10]-Annulene fits Hückel's rule,
but it's **not planar**.



[10]-annulene
10 π electrons
not aromatic

=

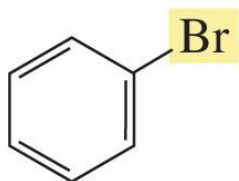


The molecule puckers to keep
these H's further away from each other.

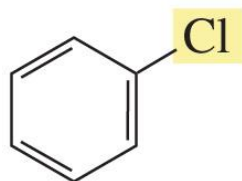
3-D representation

4.6 Nomenclature of Aromatic Compounds

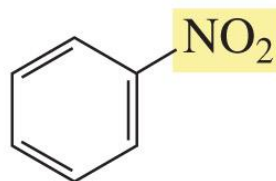
Some are named by attaching “benzene” after the name of the substituent



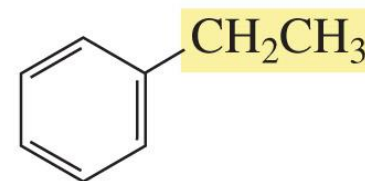
bromobenzene



chlorobenzene

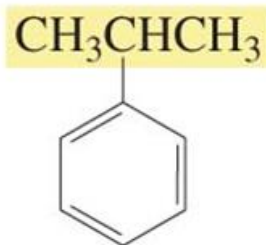


nitrobenzene
used as a solvent
in shoe polish

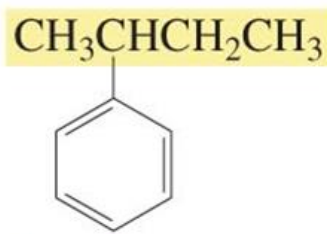


ethylbenzene

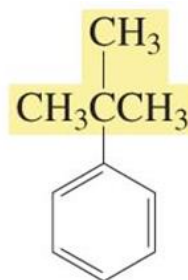
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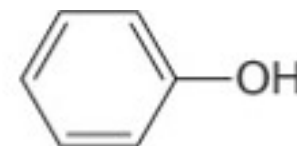
isopropylbenzene
cumene



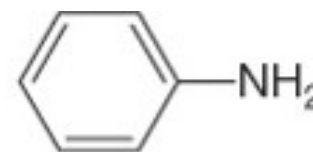
sec-butylbenzene



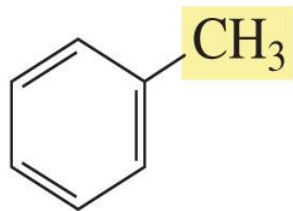
tert-butylbenzene



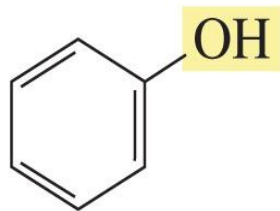
phenol
(hydroxybenzene)



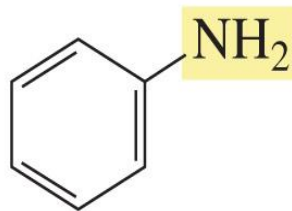
aniline
(aminobenzene)



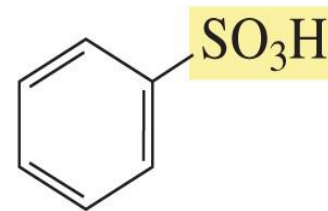
toluene



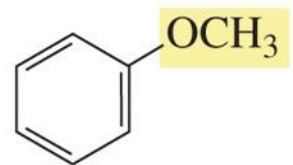
phenol



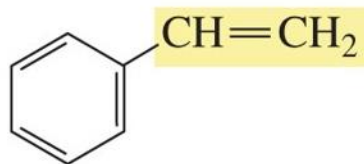
aniline



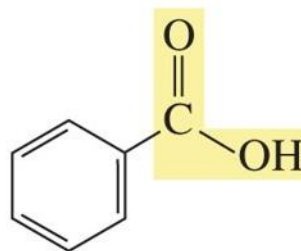
benzenesulfonic acid



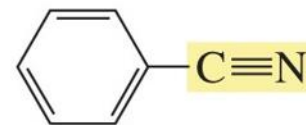
anisole



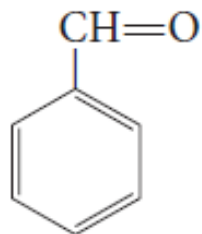
styrene



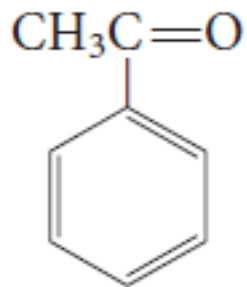
benzoic acid



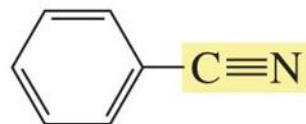
benzonitrile



benzaldehyde



acetophenone

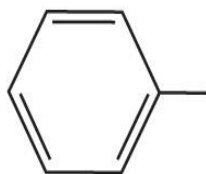
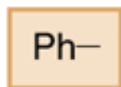


benzonitrile

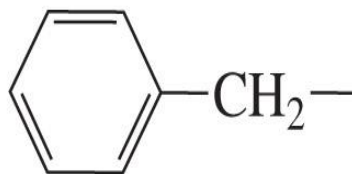
When a benzene ring is a substituent, it is called a phenyl group

A benzene ring with a methylene group is called a benzyl group

abbreviated as

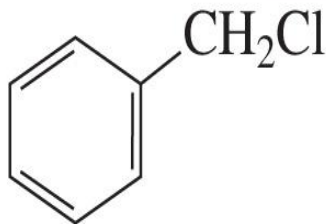


a phenyl group

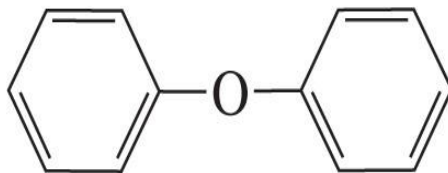


a benzyl group

- A phenyl group (C_6H_5-) is formed by removing one hydrogen from benzene (C_6H_6).



chloromethylbenzene
benzyl chloride

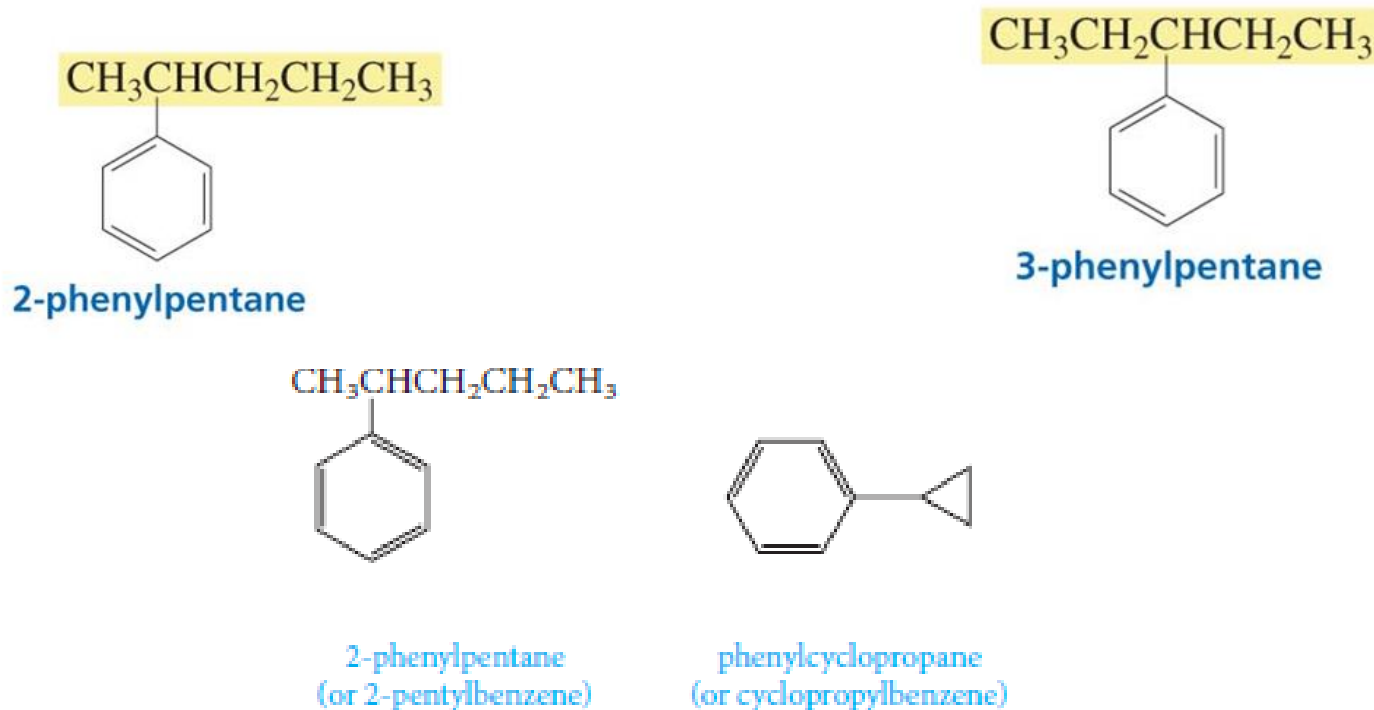


diphenyl ether

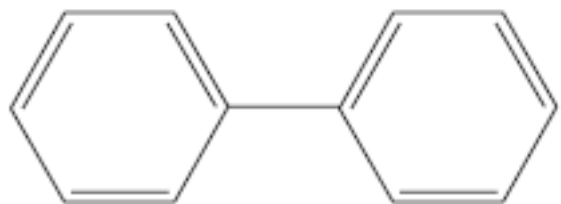


dibenzyl ether

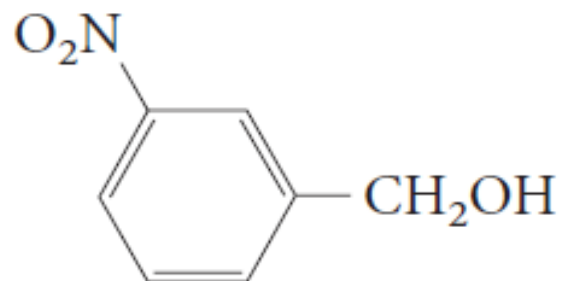
With the exception of toluene, benzene rings with an alkyl substituent are named as alkyl-substituted benzenes or as phenyl-substituted alkanes



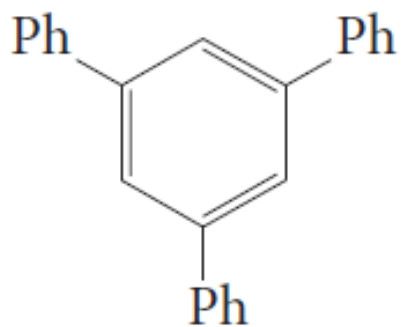
Aromatic hydrocarbons, as a class, are called **arenes**. The symbol **Ar** is used for an **aryl group**, just as the symbol **R** is used for an **alkyl group**. Therefore, the formula **Ar-R** would represent any arylalkane.



biphenyl



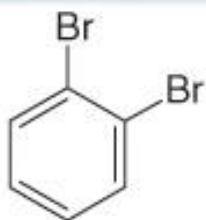
m-nitrobenzyl alcohol



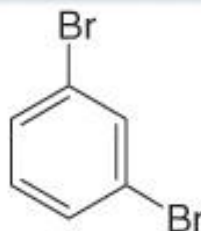
1,3,5-triphenylbenzene

In disubstituted benzenes, the relative positions of the two substituents are indicated by numbers or by prefixes

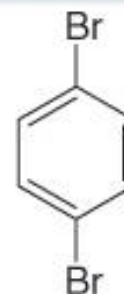
1,2-disubstituted benzene
ortho isomer



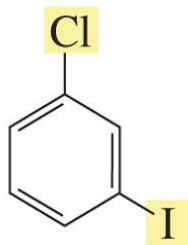
1,3-disubstituted benzene
meta isomer



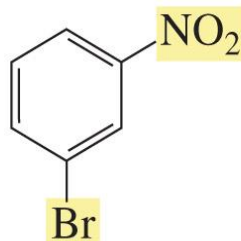
1,4-disubstituted benzene
para isomer



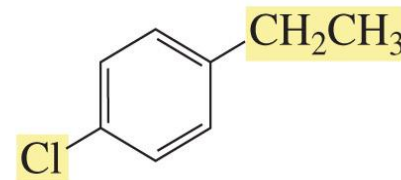
The two substituents are listed in alphabetical order



1-chloro-3-iodobenzene
meta-chloriodobenzene
not
1-iodo-3-chlorobenzene or
meta-iodochlorobenzene

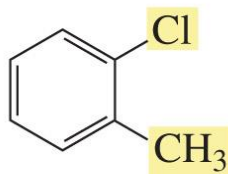


1-bromo-3-nitrobenzene
meta-bromonitrobenzene



1-chloro-4-ethylbenzene
para-chloroethylbenzene

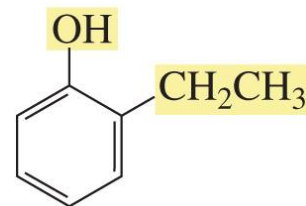
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2-chlorotoluene
ortho-chlorotoluene
not
ortho-chloromethylbenzene



4-nitroaniline
para-nitroaniline
not
para-aminonitrobenzene



2-ethylphenol
ortho-ethylphenol
not
ortho-ethylhydroxybenzene

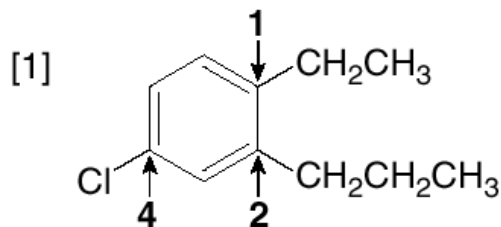
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Naming Polysubstituted Benzenes

For three or more substituents on a benzene ring:

1. Number to give the lowest possible numbers around the ring.
2. Alphabetize the substituent names.
3. When substituents are part of common roots, name the molecule as a derivative of that monosubstituted benzene. The substituent that comprises the common root is located at C1.

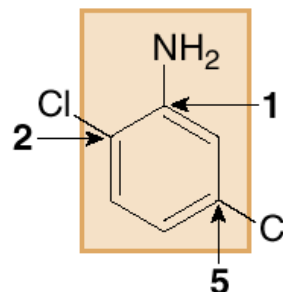
Examples of naming polysubstituted benzenes



- Assign the lowest set of numbers.
- Alphabetize the names of all the substituents.

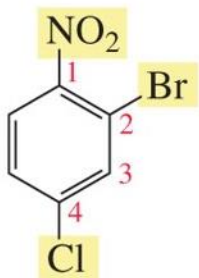
4-chloro-1-ethyl-2-propylbenzene

[2]

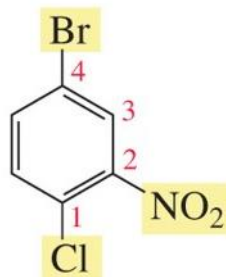


- Name the molecule as a derivative of the common root **aniline**.
- Designate the position of the NH_2 group as "1," and then assign the lowest possible set of numbers to the other substituents.

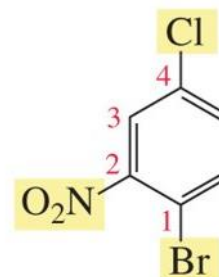
2,5-dichloroaniline



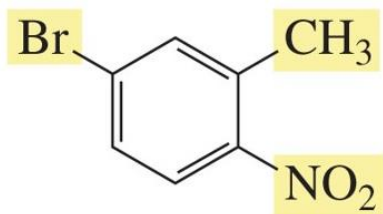
2-bromo-4-chloro-1-nitrobenzene



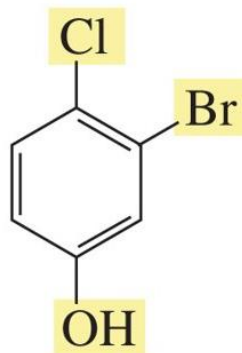
4-bromo-1-chloro-2-nitrobenzene



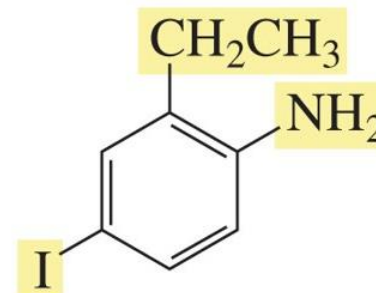
1-bromo-4-chloro-2-nitrobenzene



5-bromo-2-nitrotoluene



3-bromo-4-chlorophenol



2-ethyl-4-iodoaniline

PROBLEM 4.7 Draw the structure of

a. o-dinitrobenzene

b. m-nitrotoluene

c. p-bromophenol

d. 4-chloroaniline

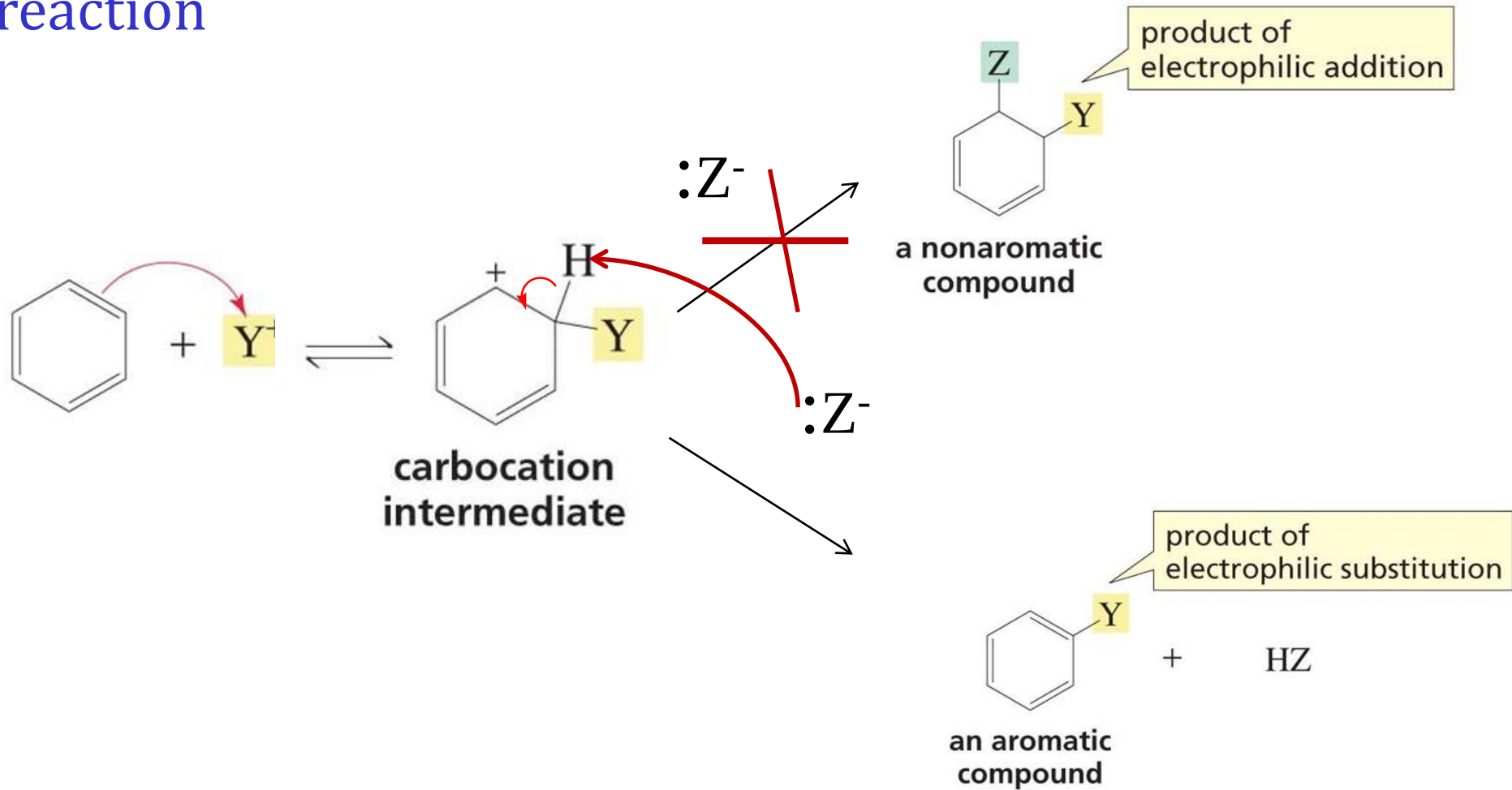
Reaction of Benzene

Aromatic compounds such as benzene undergo
electrophilic aromatic substitution reactions (EAS):

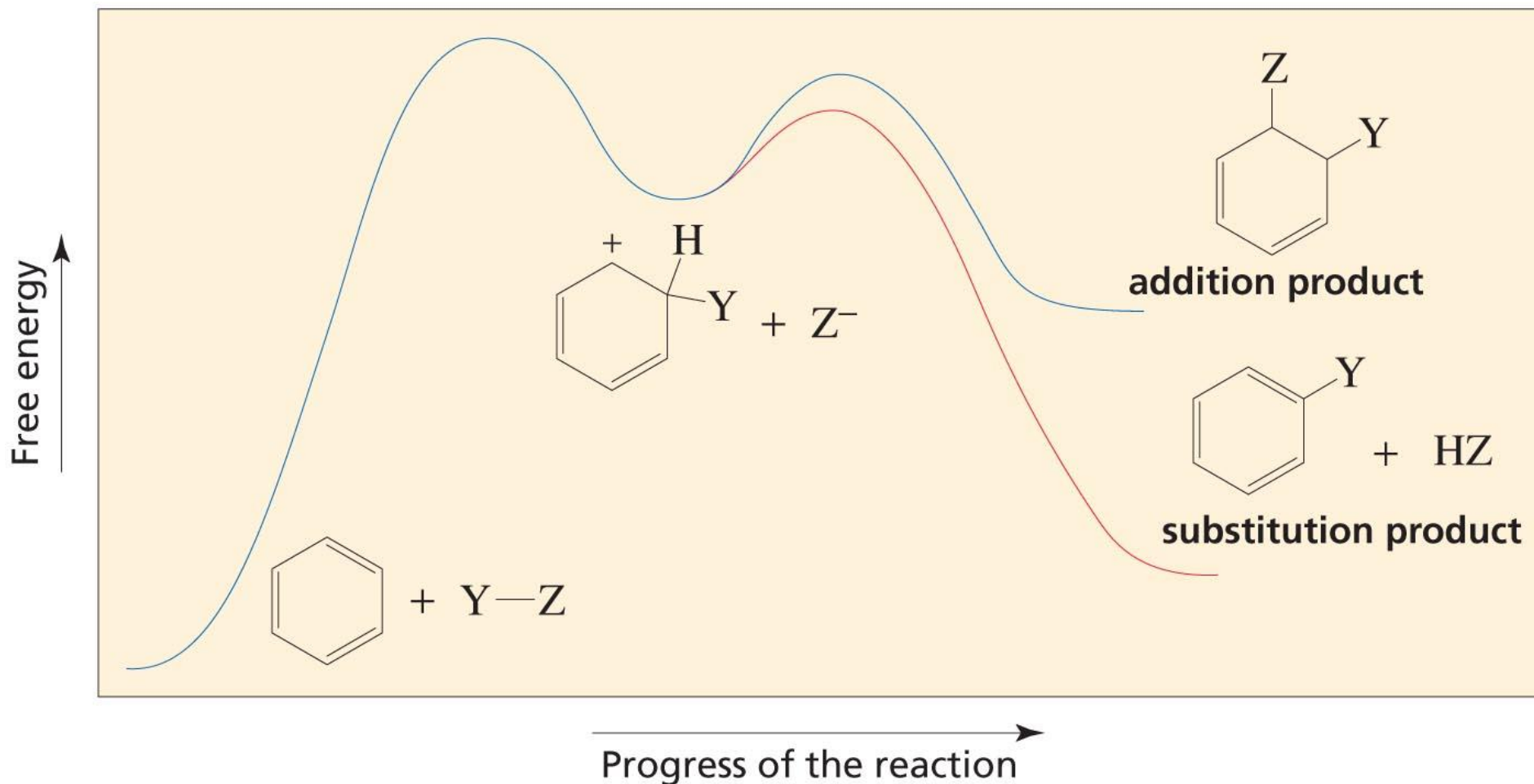


Benzene is a **nucleophile** that reacts with an electrophile

An electrophilic substitution yields an aromatic product, which is significantly more stable than the addition reaction

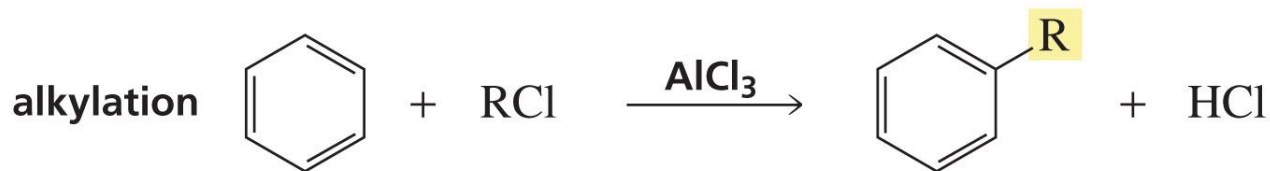
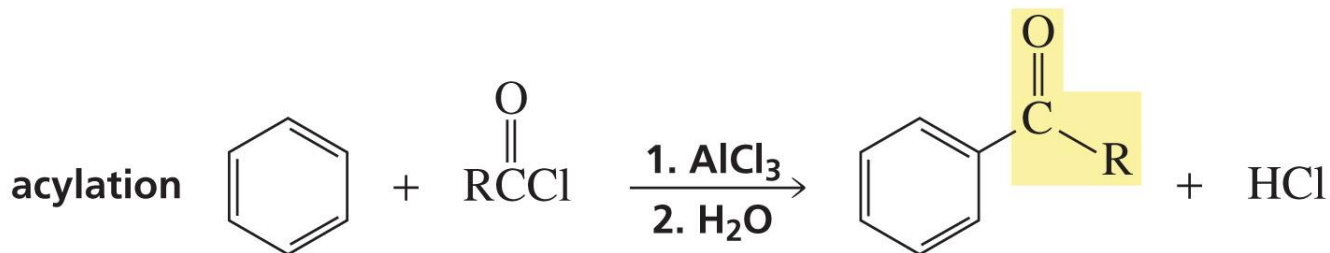
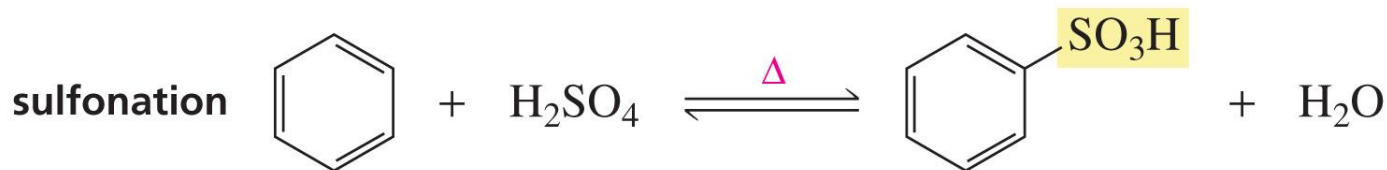
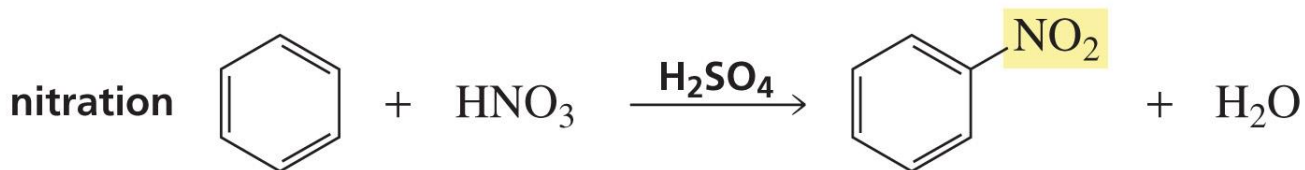
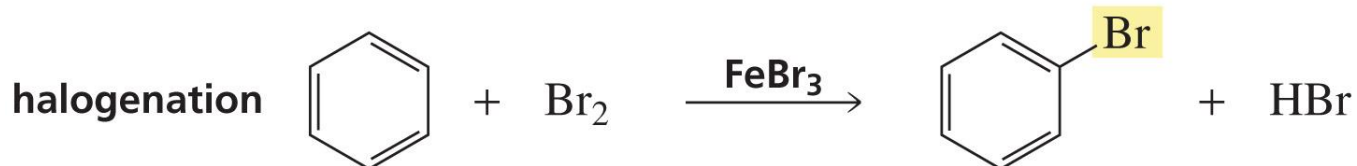


Reaction Coordinate Diagrams for the Two Benzene Reactions

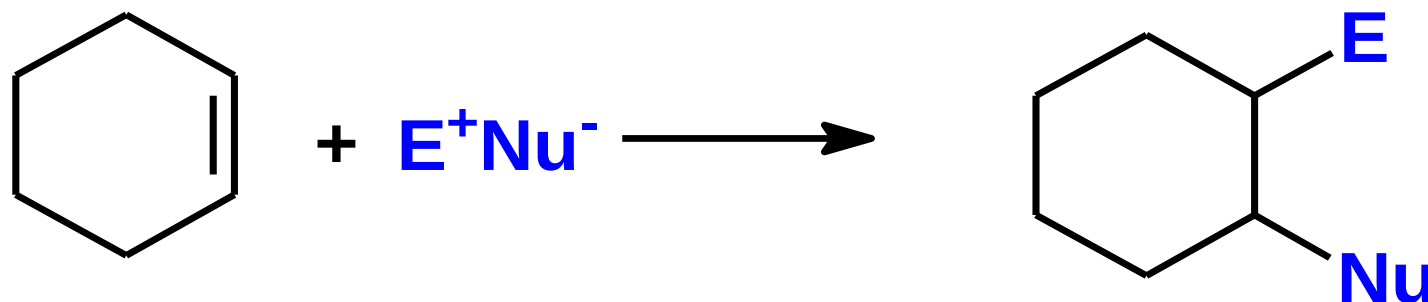
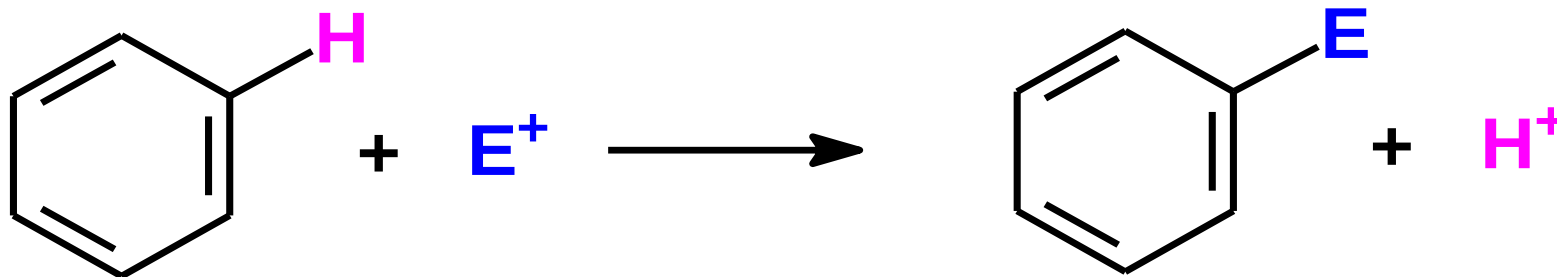


Electrophilic Aromatic Substitution Reactions

The most common reactions of aromatic compounds involve substitution of other atoms or groups for a ring hydrogen on the aromatic unit.

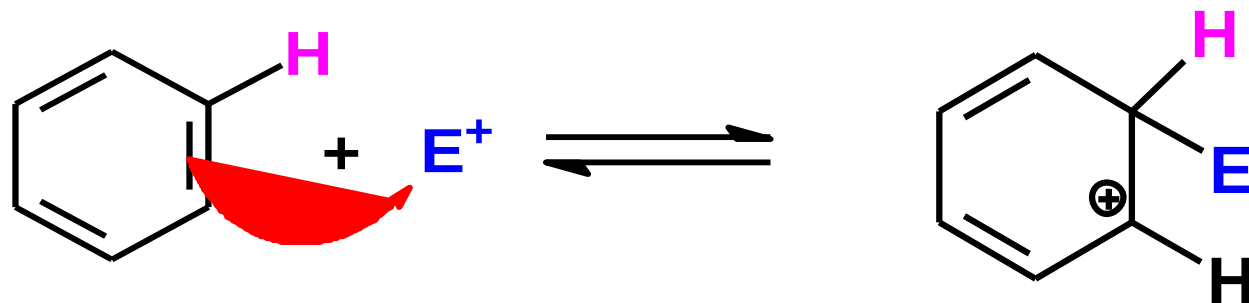


Electrophilic Aromatic Substitution



- Aromatic ring systems react with electrophiles differently than alkenes.
 - We obtain a substitution rather than an addition product.

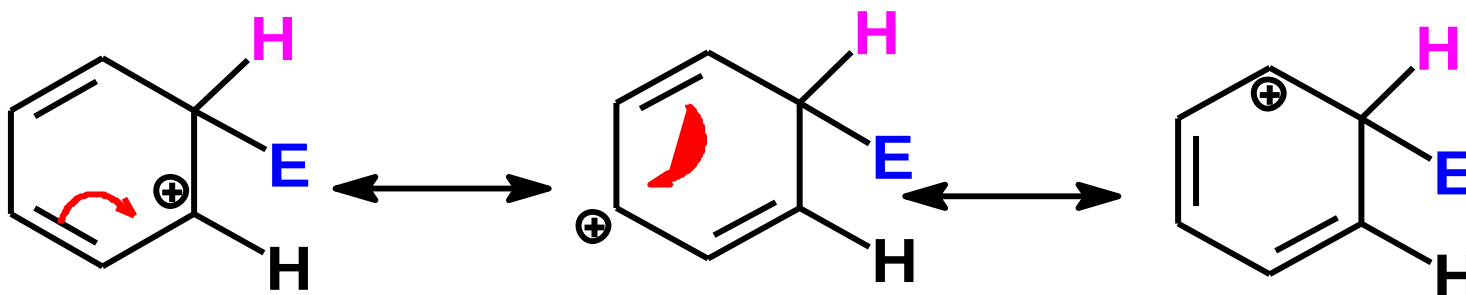
4.9 The Mechanism of Electrophilic Aromatic Substitution



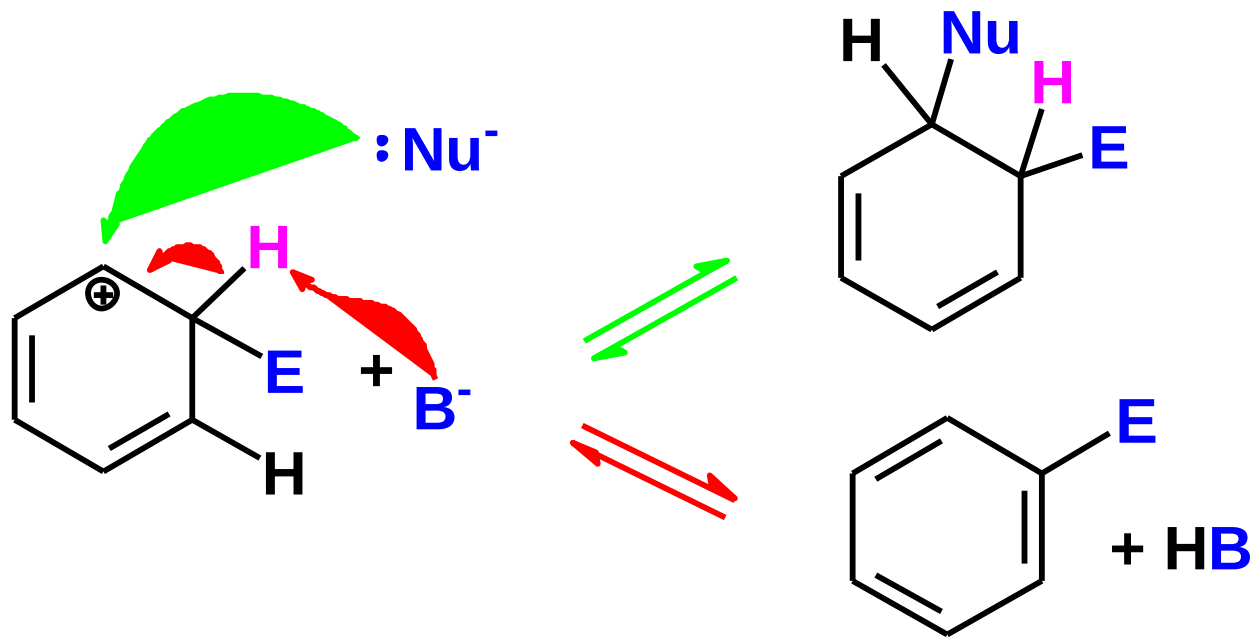
First, we get a carbocation intermediate. At the carbon that adds the electrophile, **rehybridization** ($sp^2 \rightarrow sp^3$) takes place. Molecule loses aromatic character.

Mechanism

Carbocation is a resonance stabilized structure.



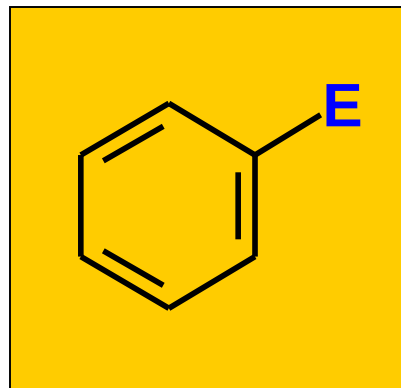
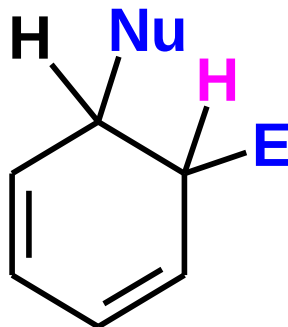
Mechanism



The electron-rich part of the reagent can act either as **nucleophile** in an addition reaction or as a **base** to remove a proton.

Mechanism

The two reactions would result in different products.



The exceptional stabilization for the aromatic compounds leads only to the substitution (addition elimination) product.

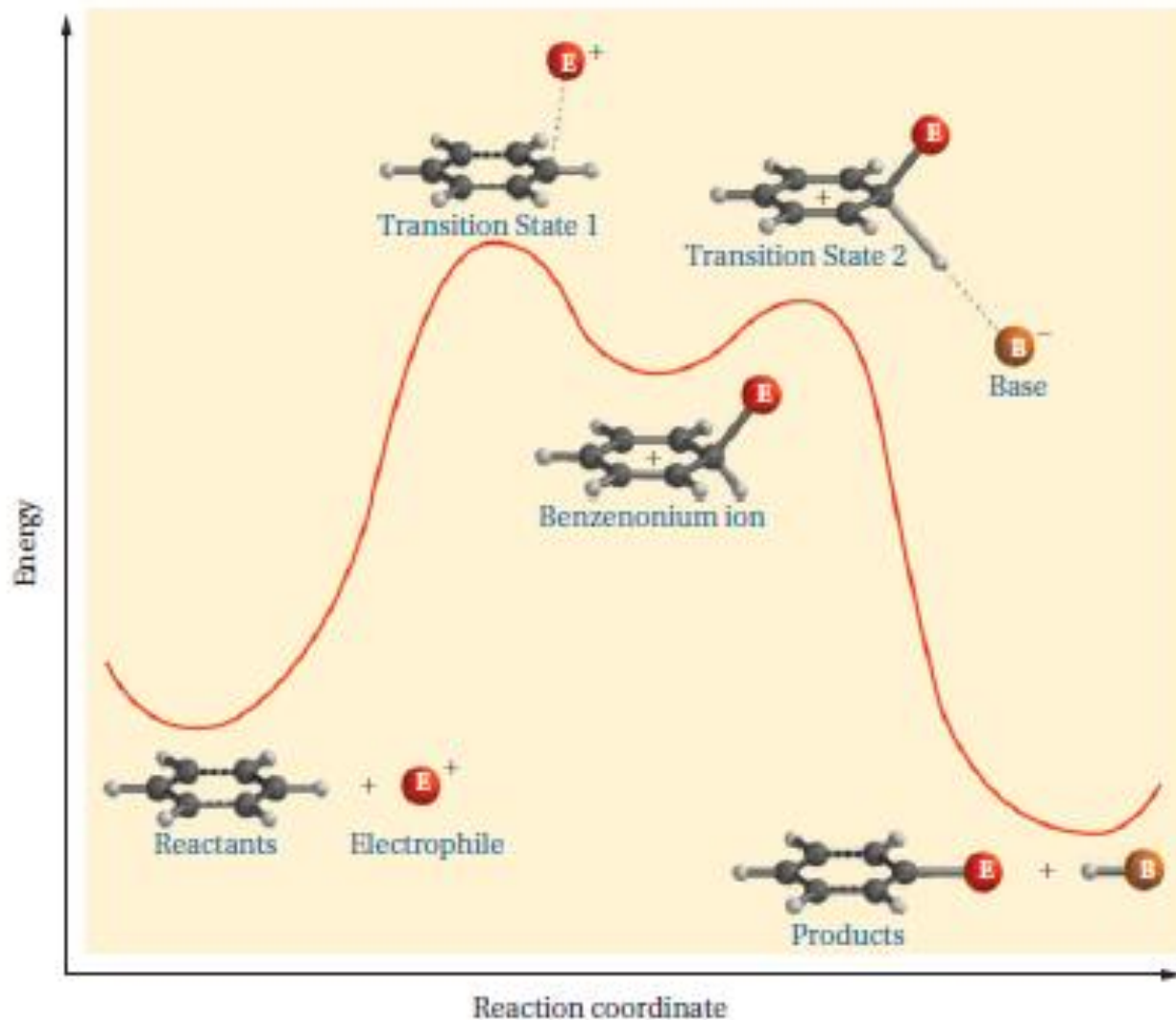


Figure 4.5

A reaction energy diagram for electrophilic aromatic substitution. The electrophile adds to benzene in an endothermic first step, with loss of aromaticity. A proton at the carbon attacked by the electrophile is lost in an exothermic second step, with restoration of aromaticity.

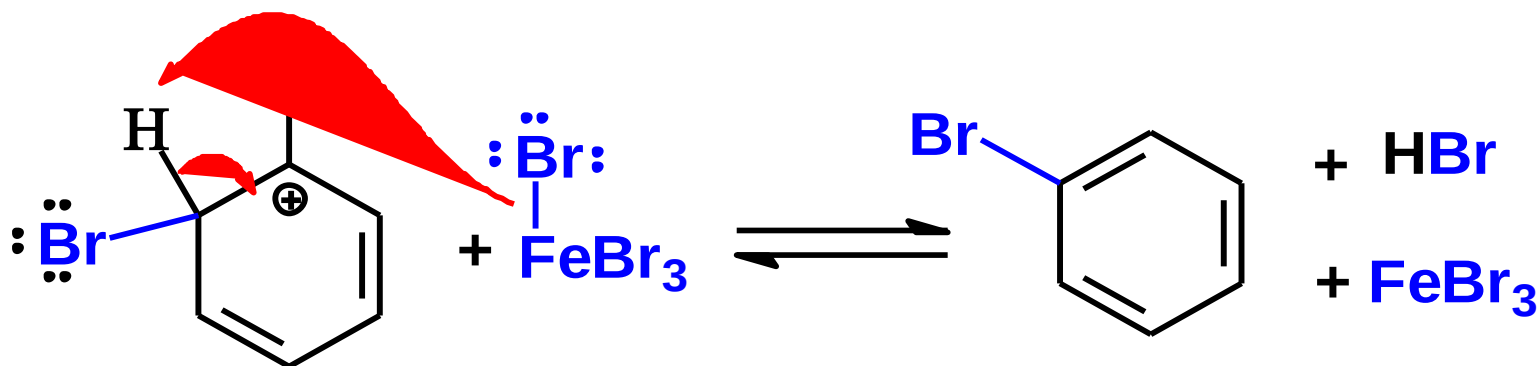
Electrophilic Aromatic Substitution

1. Halogenation:

1st step is generating an electrophile:

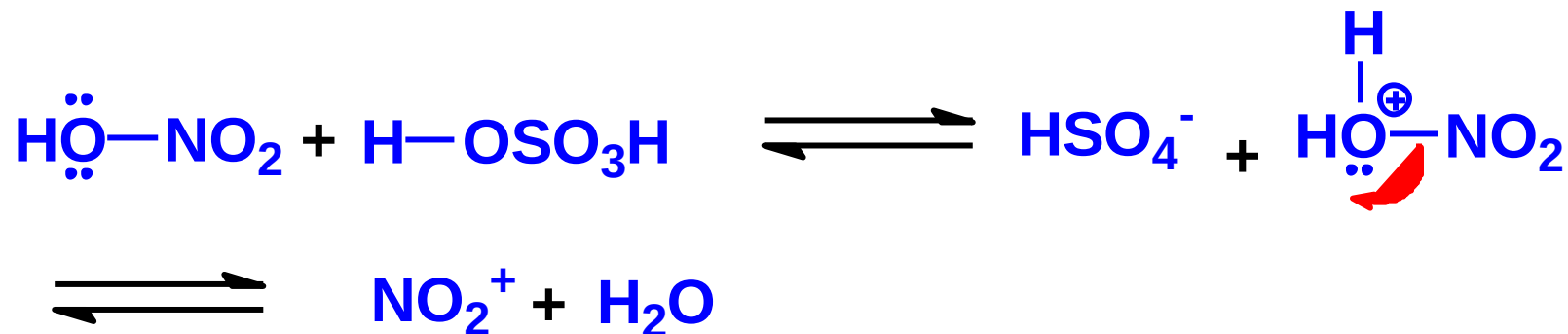


FeBr₃ is a Lewis acid that coordinates with Br₂ and polarizes it, generating an **electrophilic Br⁺**

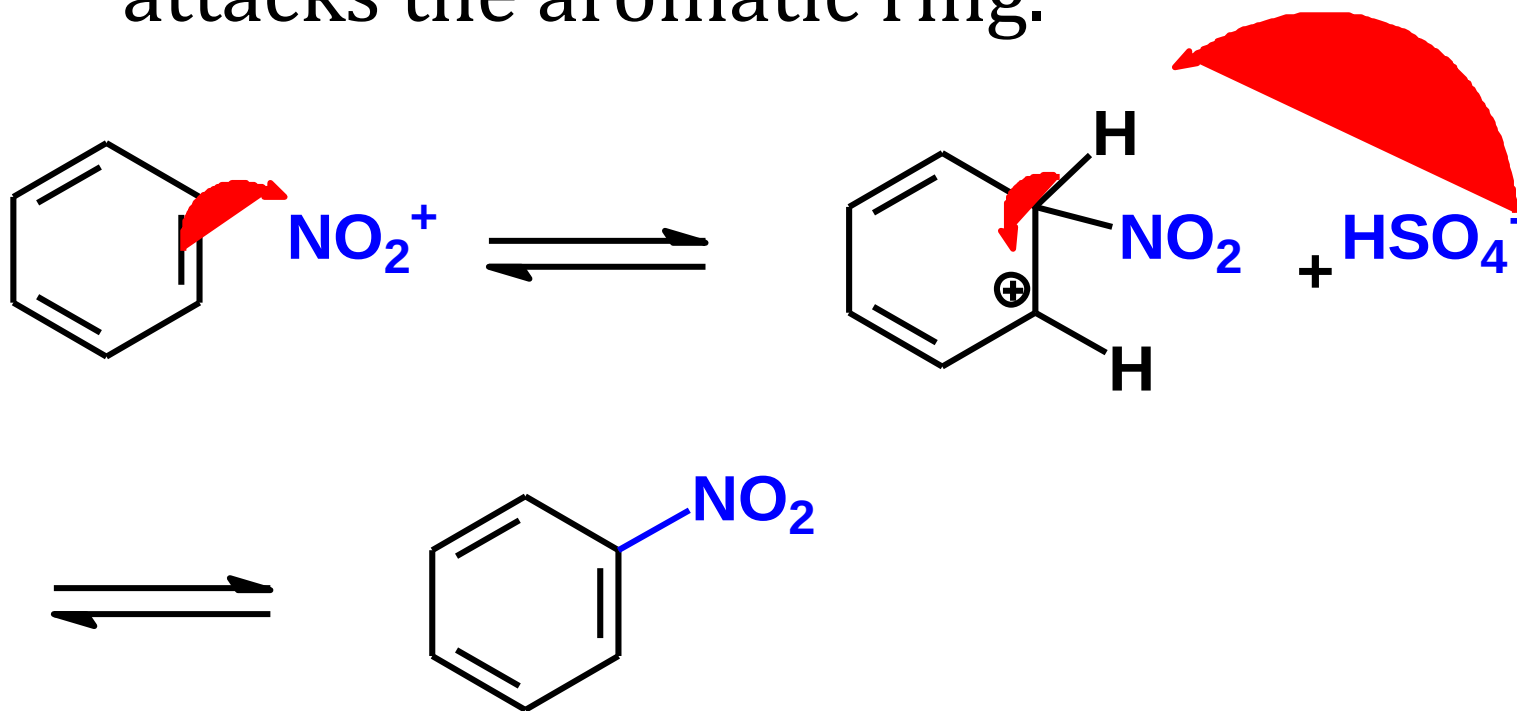


2. Nitration:

1st step is generating an electrophile, the **nitronium ion**:

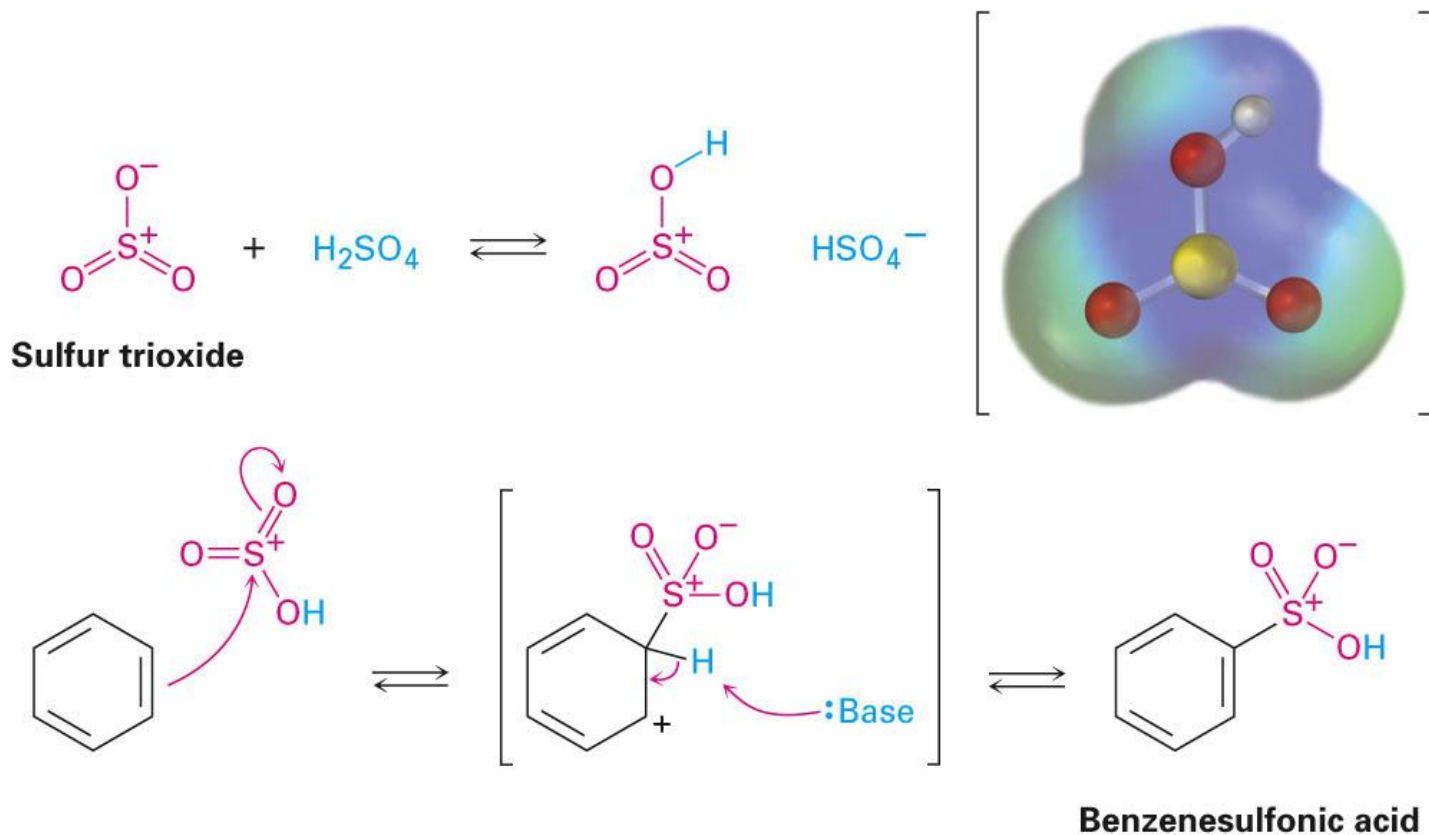


The highly electrophilic nitronium ion attacks the aromatic ring.

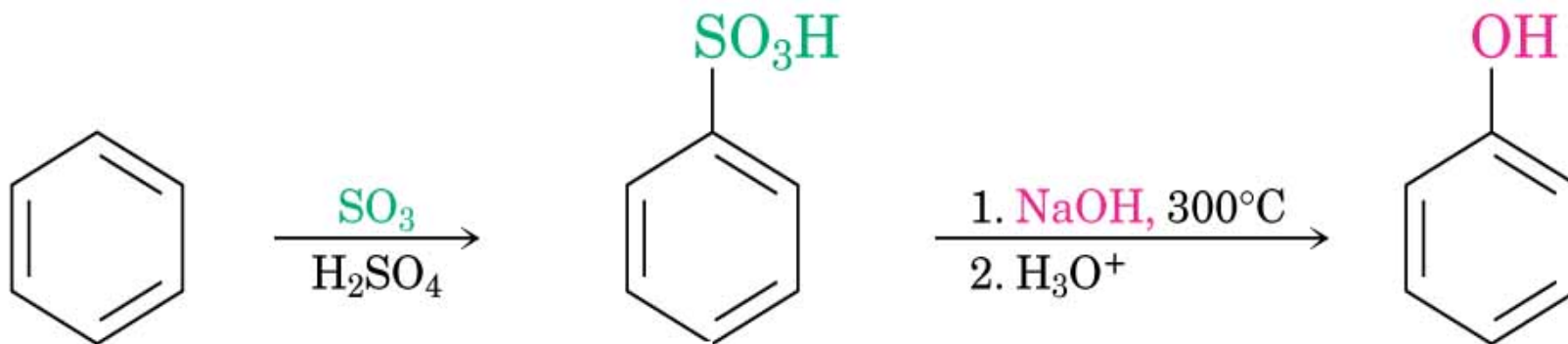


3. Sulfonation:

- Substitution of H by SO_3 (sulfonation)
- Reaction with a mixture of sulfuric acid and SO_3 ("Fuming H_2SO_4 ")
- Reactive species is sulfur trioxide or its conjugate acid



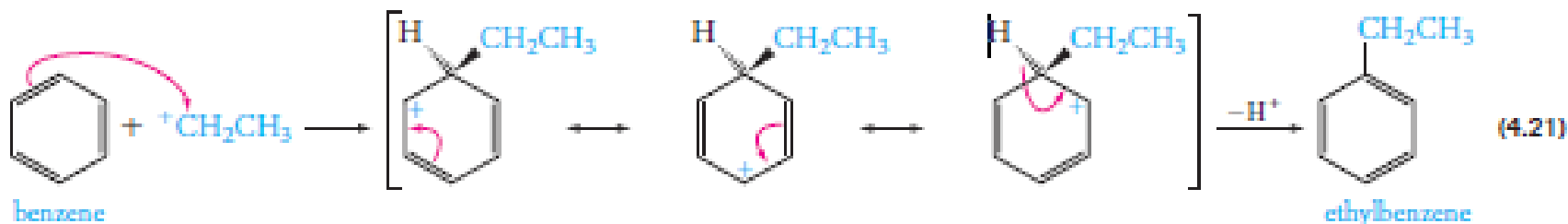
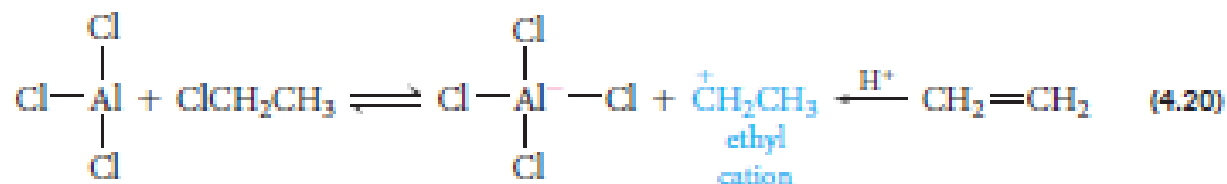
The products, sulfonic acids, are strong organic acids. Also, they can be converted to phenols by reaction with base at high temperatures.



The Friedel–Crafts Reactions

4. Alkylation of Aromatic Rings:

- Alkylation among most useful electrophilic aromatic substitution reactions
- Aromatic substitution of R^+ for H^+
- Aluminum chloride promotes the formation of the carbocation



The Friedel–Crafts Reactions

5. Acylation of Aromatic Rings

- acyl group, —COR

Similar to alkylation

- Reactive electrophile: resonance-stabilized

